



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECJ2203: Software Engineering

System Documentation (SD)

KADA Online System

Version 1

30 November 2024

Faculty of Computing

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Revision Page

a. Overview

System Documentation serves as a document that describes all aspects of a system's development, operation and maintenance. It serves as a centralized repository for information about our KADA system including the system's requirements, design and testing processes. System Documentation provides a holistic view of these three primary components which are the SRS, SDD and STD. SRS outlines the functional and non-functional requirements of our system. SDD provides information about the system's architecture including the designs, structure and data flow. STD describes the testing methodology. This document ensures clarity, quality and could improve the communication between all stakeholders, developers and operators throughout the development lifecycle.

b. Target Audience

The system documentation is designed for the project stakeholders such as the board of directors and admin users, who need to understand the system's functionalities and impact on operations. It is also intended for system developers and the quality assurance team to guide implementation and testing.

c. Project Team Members

Member Name	Role	Task	Status
Cheryl Cheong Kah Voon	Project manager	• Create user story for new user and member • Draw sequence diagram • Draw activity diagram	Partially complete
Lau Yee Wen	System analyst	• Document launch phase • Create user story for admin and board director • Draw sequence diagram • Draw activity diagram	Complete

Chau Ying Jia	System designer	• Design and plan the task • Go through the SD and get the important notes • Write the project overview and scopes	Complete
Neo Li Xin	System developer	• Draw state diagram • Draw class diagram • Draw a use case diagram	Partially complete
Woo Cheng Shuan	System testing	• List non-functional performance & other requirement • Write design constraints	Complete

d. **Version Control History**

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Cheryl Cheong Kah Voon Lau Yee Wen Chau Ying Jia Woo Cheng Shuan Neo Li Xin	Partially completed all sections in the system documentation	30/11/2024
2.0	Cheryl Cheong Kah Voon Lau Yee Wen Chau Ying Jia Woo Cheng Shuan Neo Li Xin		

Note:

This System Documentation (SD) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998), Software Design Descriptions (SDD) (IEEE Std. 1016-1998 1), and Software Test Documentation (IEEE Std. 829-2008) that are simplified and customized to meet the need of SECJ2203 course at School of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly.

Table of Contents

1. Introduction	1
1.1 Purpose	2
1.2 Scope	2
1.3 Definitions, Acronyms and Abbreviations	3
1.4 References	3
1.5 Overview	4

1. Introduction

1.1 Purpose

System Documentation provides an overarching framework that integrates the SRS, SDD, STD, and other relevant materials, offering a full lifecycle view of the system. SD serves as a central repository of information which includes the system's functionality, design, requirements and constraints enabling the administrators and the board of directors of KADA to understand the scope of the system. SD provides a development blueprint that allows developers to understand the architectural framework and functional workflows of the system, as well as outlines the functional and technical details for system design and implementation. SD also provides detailed testing procedures and user requirements to ensure the system meets predefined specifications and quality benchmarks. SD helps to troubleshoot and maintain the system by executing and validating test cases with some new users during the implementation process. SD ensures that all parties have access to relevant and actionable information.

1.2 Scope

Our software product is the KADA System. It is a website designed to automate and streamline KADA's internal processes, particularly used in member registration and loan applications. This system aims to reduce manual work, improve operational efficiency and enhance usability through a digital solution. The KADA system provides a combined platform for administrators, members, and the board of directors, with core functionalities such as member management, loan application and processing, and data analytics and reporting.

1. Member Registration

The new user can register their membership by submitting the member application online to reduce the burden of manual data handling and processing.

2. Loan Application

The loan application can also be submitted online through the system.

3. Application status checking

This feature can allow the applicant to check the real-time status of their application through the website. Once the application is approved, they will receive an automated email or SMS notification.

5. Check monthly and annual report

The system will generate monthly and annual reports for the members from the database to allow members to know their financial performance from time to time. The system will also generate an annual report for the board of directors to offer a comprehensive overview

1.3 Definitions, Acronyms and Abbreviations

This section provides the definition of fall terms, acronyms and abbreviations used in this System Documentation.

- | | |
|---------|--|
| 1. KADA | Lembaga Kemajuan Pertanian Kemubu |
| 2. SD | System Documentation |
| 3. SRS | System Requirements Specification |
| 4. SDD | System Design Document |
| 5. STD | System Test Document |
| 6. CCTV | Closed-circuit Television |
| 7. ISO | International Organization for Standardization |
| 8. TLS | Transport Layer Security |
| 9. SSL | Secure Socket Layer |
| 10. CPU | Central Process Unit |
| 11. RAM | Random Access Memory |

1.4 References

[1] Visual Paradigm, “What is Class Diagram?,” *Visual Paradigm Guides*. [Online].

Available:

<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-class-diagram/>.

[2] GeeksforGeeks, “Unified Modeling Language (UML) State Diagrams,” *GeeksforGeeks*. [Online]. Available: <https://www.geeksforgeeks.org/unified-modeling-language-uml-state-diagrams/>.

1.5 Overview

In this System Documentation, we will need testers to test that the system satisfies our design or technical requirements. We first introduce the five personas we are going to have throughout this section to let our development team have a clear understanding of the needs of users and to whom the system is designed to ensure that the system is well-developed. Next, we are going to explain our system features using a use case diagram, description, class diagram and state diagram. We also include a clear flow of how each persona used our system and what function they are using. We mainly focus on the user story for each use case and the process as there are different scenarios of user story. We are using sequence diagrams to help us to have a clearer view in this section. Lastly, we will finalize our research and make a conclusion on the aspect of system performances, requirements and design constraints.

2. Specific Requirements

2.1 Persona

In this section, we will be using a persona to identify the characteristics of the users when meeting certain scenarios. These observations will allow us to make a better judgement about the system's functionality and design. Below are the personas we identified.

2.1.1 Persona 1 (User type 1/Actor 1)

User Need

A secretary managing membership applications.

User Stories

Sarah has been the secretary at KADA for over five years. Her main responsibility includes managing membership applications. Every week, she receives a stack of paper forms from potential members. Some are handwritten and barely legible, others are missing information, and a few arrive late or are misplaced altogether.

She spends hours manually sorting, verifying, and cross-checking details with incomplete spreadsheets. On a hectic day, Sarah needs to catch up on time, stuck between phone calls from applicants asking for updates and trying to find missing files in a growing pile on her desk. By the end of the day, her frustration peaks when she realises she accidentally approved a duplicate application and missed a deadline for another.

Sarah dreams of having a digital system where applicants can easily submit forms, validate their information, and track their application status. This would save her time, reduce errors, and allow her to focus on more strategic tasks. She wishes for a solution that's simple, efficient, and user-friendly—not just for her, but for the applicants as well.

2.1.2 Persona 2 (User type 2/Actor 2)

User Need

An archive staff ensures forms are stored securely and easily accessible.

User Stories

As a staff of KADA, Jonathan is tasked with managing and storing the physical forms in the organisation's archive. He has to make sure that the forms are stored in their respective places so that they are easily accessible by other staff.

One day Jonathan notices that part of the forms are missing. He tried asking other staff about the missing forms but to no avail, no one knew where they were. Confused and panicked, he checked the CCTV footage and confirmed that one of the staff did indeed steal the forms. Disappointed by the incident Jonathan reported it to the organization's management.

After acknowledging the potential security risks, Jonathan proposed to install a digital system where the forms can only be accessed online by authorized users. This system offers better management and security to the customer's information by introducing secure login features, thus reducing the risk of unauthorized users breaching their privacy.

2.1.3 Persona 3 (User type 3/Actor 3)

User Need

A customer service staff receiving necessary financial documents.

User Stories

Judie works in the customer service department. Every day at the counter she handles the physical forms submitted by the customers. Afterwards, the customers are required to submit their bank statements themselves as part of the loan application process. These bank statements are then sent to the management department for further validation.

As time went on, Judie noticed that some customers failed to bring the necessary documents to the next appointed date. This issue causes a delay in the validation process due to incomplete information provided. It also leads to customer dissatisfaction as their mistakes cost them a lot of time.

To solve this issue Judie discussed with the IT staff to develop an online system that allows customers to access and submit their bank statements digitally. The system will maintain comprehensive and up-to-date financial status, providing members with detailed insights into their investments, loans, and other financial activities.

2.1.4 Persona 4 (User type 4/Actor 4)

User Need

An archive staff manager struggles to find space for paper storage.

User Stories

Kumar has worked as an archive staff manager for the past decade. He has been managing the physical reports and constantly checks on their organization, accuracy, and accessibility to ensure all documents are properly maintained and readily available when needed.

Back then, physical forms were much easier to handle with the help of other staff members. However, recently Kumar noticed that the increasing amount of paperloads each year are getting harder to manage. As applications increase, archive storage decreases as well as increasing workload for the staff members, causing fatigue and stress.

To address this issue, Kumar suggested implementing an online system for storing archives which would reduce manual labor, paper storage, and physical archiving while increasing productivity. This method also offers improved scalability in the future as the number of applications increases.

2.1.5 Persona 5 (User type 5/Actor 5)

User Need

The board of directors manages the report.

User Stories

Ali is the board of directors of KADA Corporation Sdn Bhd. He is responsible for managing all types of reports, including member applications, loan applications and transaction records. The managing process involves viewing, checking, validating and printing. This time-consuming approach prompts Ali to seek a centralized and efficient solution for handling all reports.

One afternoon, during a particularly hectic day, Ali receives an urgent call from the CEO demanding a full account of loan approval. As he attempts to complete the report, he sees how much time is spent searching for papers, manually checking data, and ensuring everything is accurate. Frustrated, he thinks of a way to solve this issue.

A few days later, Ali thought of creating a dashboard that could access different types of reports in real-time. The system should display the reports, export them (PDF, Excel) and print a physical copy. Not to mention a feature that can automatically validate the reports to reduce human errors and also save time.

2.2 System Features

The objective of this KADA project is to design a system encompassing four key processes: member registration, loan application, reporting, and savings management. The system provides a means for users to do their membership and loan applications can be done online. It replaces the current manual processes, which are time-consuming, prone to errors, and heavily reliant on human resources. The system also ensures better management and security systems to optimize operations while ensuring data integrity and protection.

The system features are illustrated in Figure 1.1 below. The detailed description of each module and function is tabulated in Table 1.1.

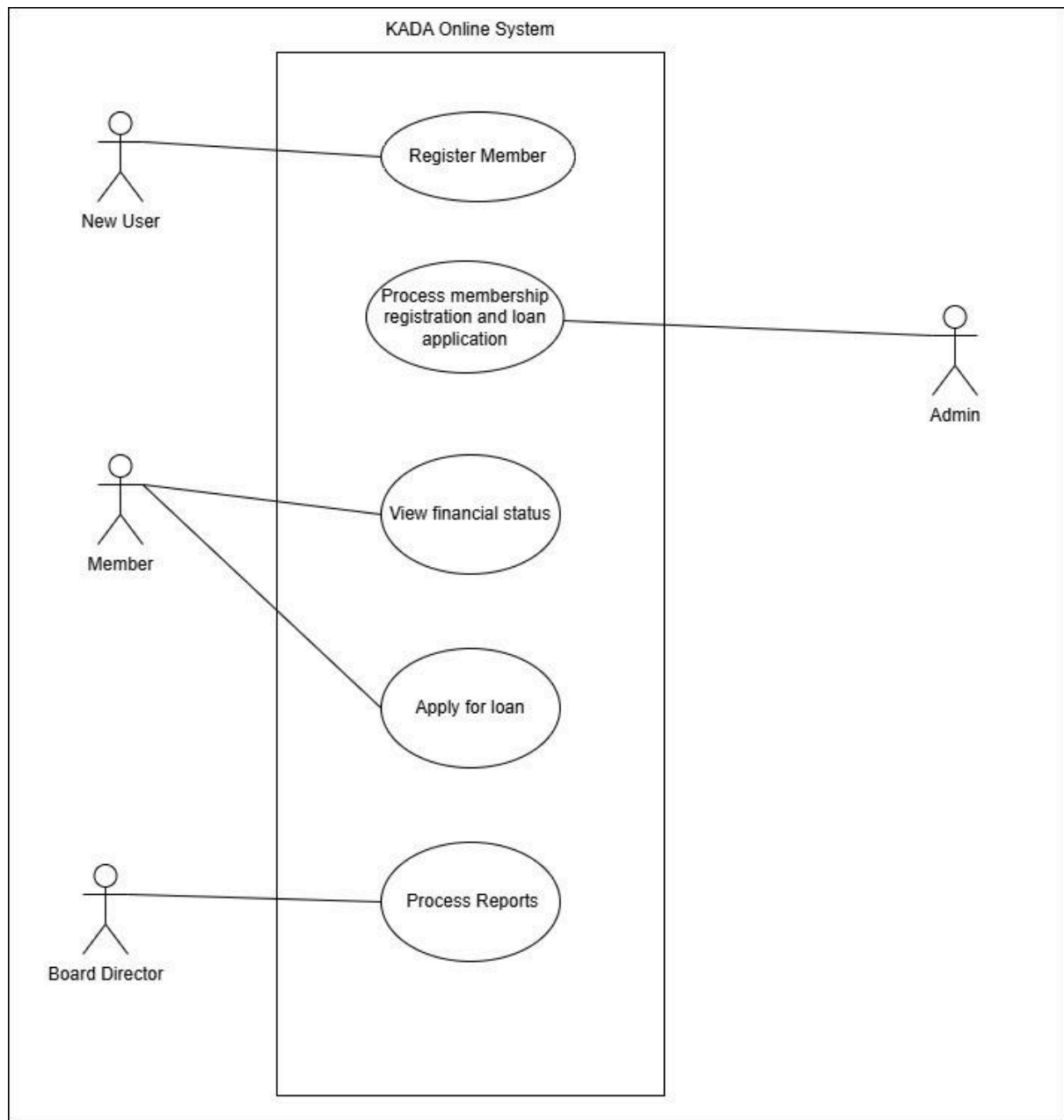


Figure 1.1: Use Case Diagram for KADA Online System

Table 1.1: Description of Module and Functions for KADA Online System

Function	Description
Register member	Allow new user to register as a member of KADA
Process membership registration and loan application	Admin will manage the membership and loan application forms, ranging from reviewing and verifying applications based on set criteria
View financial status	Check recent transactions and financial account
Apply for loan	Allow members to register for loan application
Process reports	Read and validate the online forms

Figure 1.2 shows the class diagram of the entire member application and loan application process of the KADA online system. It represents the relationships between each class and shows how their attributes interact among different classes.

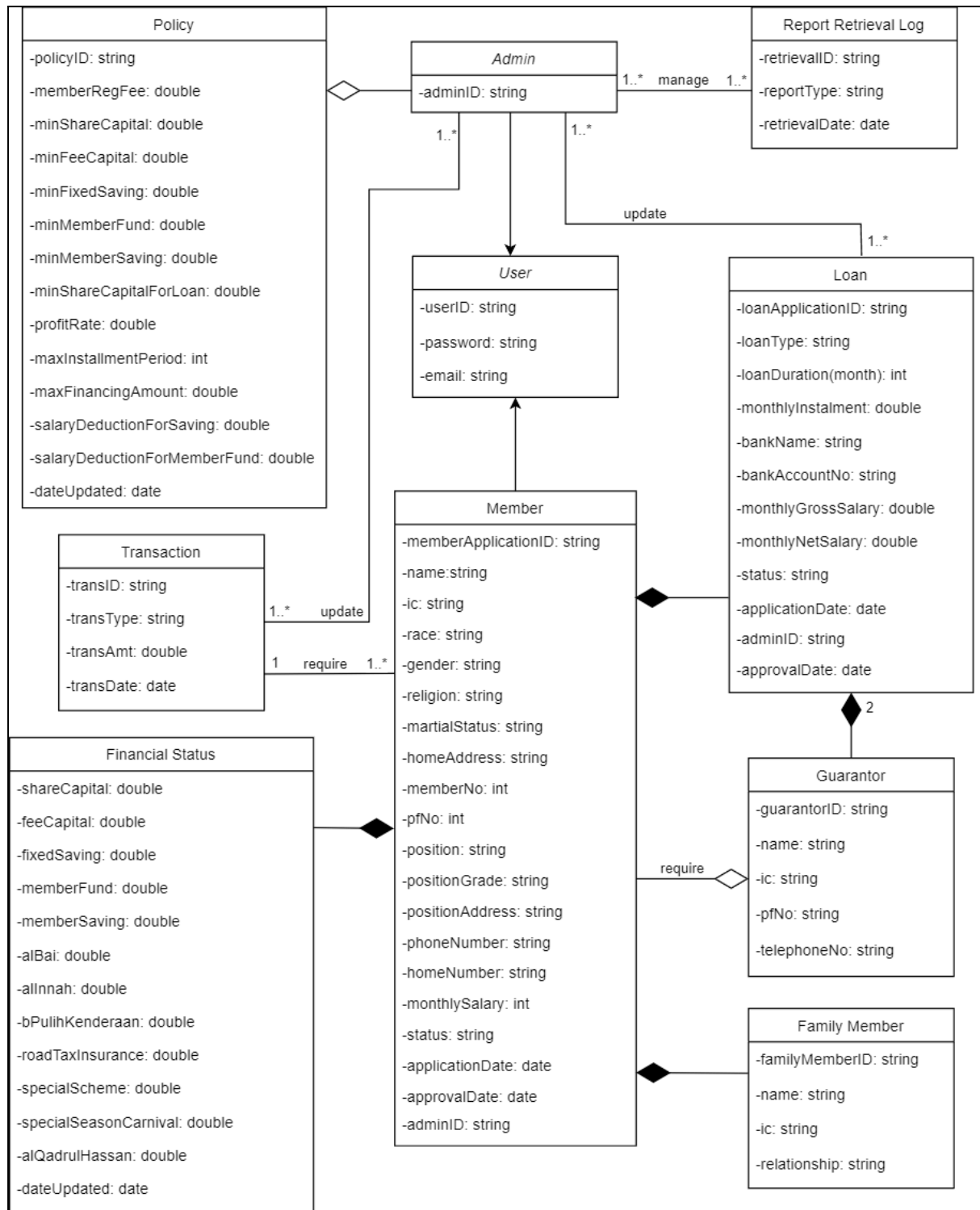


Figure 1.2: Domain Model for Online KADA System

User Class

Represents all users that use the system. The information in this class will act as login credentials to the system.

- userID: A unique identifier for user
- password: Password to login system
- email: User email account

Member Class

Represents a person registered in the system as a member. As a user applies for a member, an object of the member is created.

- memberApplicationID: A unique identifier for membership application.
- name: Name of the member.
- ic: Identity card (MyKad) number of the member.
- gender: Gender of the member. This attribute is an enumeration with the values Male (Lelaki) and Female (Perempuan).
- religion: Religion of the member. This attribute is an enumeration with the values Islam, Buddhism (Buddha), Hinduism (Hindu), Christianity (Kristian) and Others (Lain-lain).
- race: Race of the members. This attribute is an enumeration with the values Malay (Melayu), Chinese (Cina), Indian (India) and Others (Lain-lain).
- maritalStatus: Marital status of the member.
- homeAddress: Address of the member. This includes line1, line2, Postcode (Poskod), City (Bandar) and State (Negeri).
- memberNo: Member number of the member. The member number would be available as the application is approved.
- pfNo: PF number of the member.
- position: Position held by the member.
- positionGrade: Position grade of the member.
- officeAddress: Address of the member on duty. This includes line1, line2, Postcode (Poskod), City (Bandar) and State (Negeri).
- phoneNumber: Phone number of the member.
- homeNumber: Home number of the member.
- monthlySalary: Monthly salary of the member with the currency of RM.
- status: Application status of the member. This attribute is an enumeration with the values Processing (Sedang Diproses), Rejected (Ditolak) and Accepted (Diterima).
- applicationDate: Date when the application is submitted.

The Member class is inherited from the User class. A member whose status is accepted would be eligible to apply for a loan.

Financial Status Class

Represents the financial status of the associated member.

- shareCapital: Share capital (Modah Syer) of the associated member with the currency of RM.
- feeCapital: Fee Capital (Modal Yuran) of the associated member with the currency of RM.
- fixedSaving: Fixes Saving or Fixed Deposit (Simpanan Tetap) of the associated member with the currency of RM.
- memberFund: Member Fund (Tabung Anggota) of the associated member with the currency of RM.
- memberSaving: Member Saving (Simpanan Anggota) of the associated member with the currency of RM.
- alBai: The loan amount in the Al-Bai category for the associated member in RM.
- alInnah: The loan amount in the Al-Innah category for the associated member in RM.
- bPulihKenderaan: The loan amount in the Baik Pulih Kenderaan category for the associated member in RM.
- roadTaxInsurance: The loan amount in the Road Tax & Insurance category for the associated member in RM.
- specialScheme: The loan amount in the Special Scheme (Khas) category for the associated member in RM.
- specialSeasonCarnival: The loan amount in the Special Season Carnival (Karnival Musim Istimewa) category for the associated member in RM.
- alQadrulHassan: The loan amount in the Al-Qadrul Hassan category for the associated member in RM.
- dateUpdated: The last date of the financial status is updated. The update is made by the admin and detailed in the Transaction class.

The Financial Status class is a composition of the Member class. The relationship is a one to one relationship.

Loan Class

Represents the loan application applied by the associated member.

- loanApplicationID: A unique identifier for the loan application.
- loanType: The type of loan applied by the member. This attribute is an enumeration with the values Al-Bai, Al-Innah, Baik Pulih Kenderaan, Road Tax and Insurance, Special Scheme (Khas), Special Season Carnival (Karnival Musim Istimewa) and Al-Qadrul Hassan.
- appliedLoan: The amount of loan applied by the member in RM.
- loanPeriod: The instalment period of the applied loan.
- monthlyInstalment: The monthly instalment amount of the applied loan in RM.
- bankAccountNo: The bank account number of the applicant.
- bankName: The bank associated with the bank account number provided.
- monthlyGrossSalary: The monthly gross salary of the applicant in RM.
- monthlyNetSalary: The monthly net salary of the applicant in RM.
- signature: The reference to the uploaded acknowledgement form with the signature of the applicant's employer.

- status: Application status of the member. This attribute is an enumeration with the values Processing (Sedang Diproses), Rejected (Ditolak) and Accepted (Diterima).
- applicationDate: Date when the application is submitted.

A member of the cooperative fulfilling certain criterias would be able to apply for loans.

(tbc..)

Figure 1.3 shows the state diagram for the member application process. After submission, it will be checked and validated by the admin. When successful, a report will be generated by the board of directors and the status of the new user will be updated.

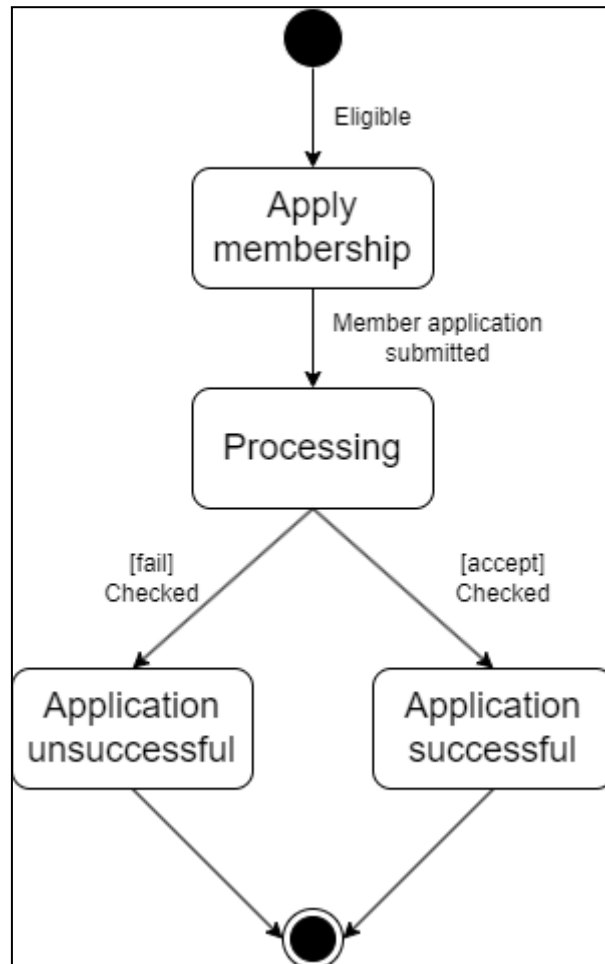


Figure 1.3: State Machine Diagram for Member Application of Online KADA System

Figure 1.4 shows the state diagram for the loan application process. During this process, the person applying must be a KADA member first and foremost. After the loan application submission, it will be checked and validated by the admin. When successful, a report will be generated by the board of directors and the status of members will be updated.

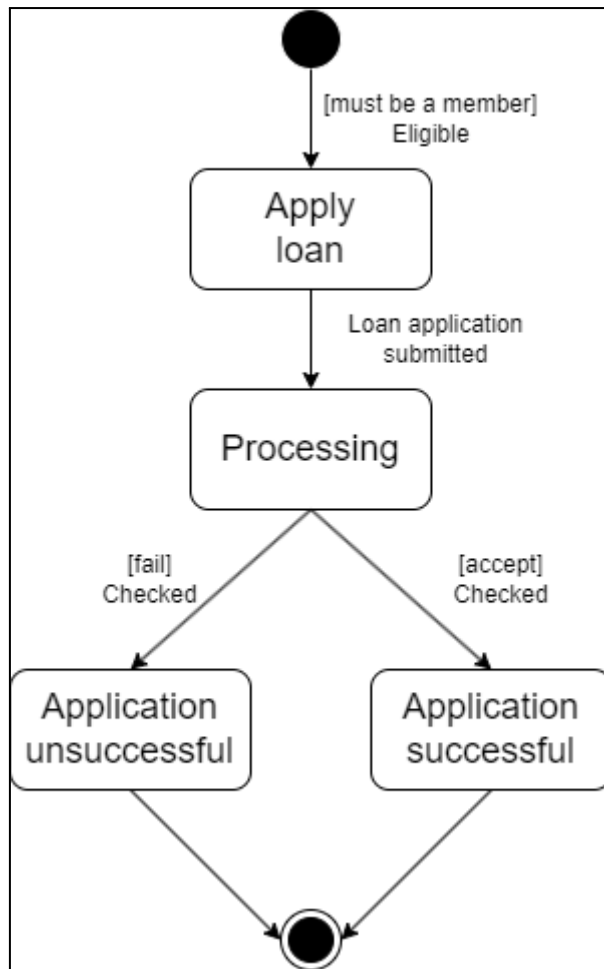


Figure 1.4: State Machine Diagram for Loan Application of Online KADA System

2.3 Launch Phase

The project's components and modules are listed in the product backlog according to priority. Table 1.2 shows the product backlog for the Kada System.

Sprint	Use Case Involved	Team Members Assigned
<i>Sprint #1</i> <i>User Module</i>	UC001 Register Member	Lau Yee Wen
<i>Sprint #2</i> <i>Member Module</i>	UC002 View financial status UC003 Apply for a loan	Cheryl Cheong Kah Voon Neo Li Xin
<i>Sprint #3</i>	UC04 Process membership registration and loan application	Chau Ying Jia

<i>Admin Module</i>		
<i>Sprint #4</i>	UC05 Process Reports	Woo Cheng Shuan
<i>Board Director Module</i>		

2.4 User Story Details

This section outlines users' goals and expectations for the KADA system, with a focus on the required actions. Each user story defines the scope, event flow, and supporting diagrams (sequence and activity) to ensure that the system meets the user's needs effectively and efficiently.

2.4.1 US001 User Story Apply for Membership

Table 2.1: User Story Description for Apply for Membership

User story: Apply for Membership
ID: US001
User Story Description: As a KADA employee I want to register as a member of the cooperative So that I can enjoy more benefits that are provided by the cooperative.
Flow of events: 1. The employee accesses the system by searching the website. 2. The member chooses the “Permohonan Anggota ” on the navigation bar. 3. The employee fills in all the personal details. 4. The employee clicks the submit button to submit the application form. 5. The system will prompt a message to confirm the submission.
Alternative flow: 1. If the personal information is incomplete, the system will prompt a message to remind the employee to fill in the incomplete place. 2. If the application is rejected, the employee must reapply and cannot log in to the system.
Acceptance Criteria

Precondition:

1. The employee must be a current employee of the KADA.

Postcondition:

1. If the application is approved, the member will receive a notification and the employee can log into the system to enjoy the benefits provided by the cooperative.
2. If the application is rejected, the employee will receive a notification. The employee must reapply and cannot log in to the system.

Other conditions:

1. The new member details will be stored in the member database.
2. A rejected application must clearly notify the reason for the rejection so that the employee can easily reapply.

Exception flow:

1. If the system is down, the system will prompt a message to advise the employee to try later.
2. If the employee has invalid details, the system will prompt a message to remind the employee to re-enter

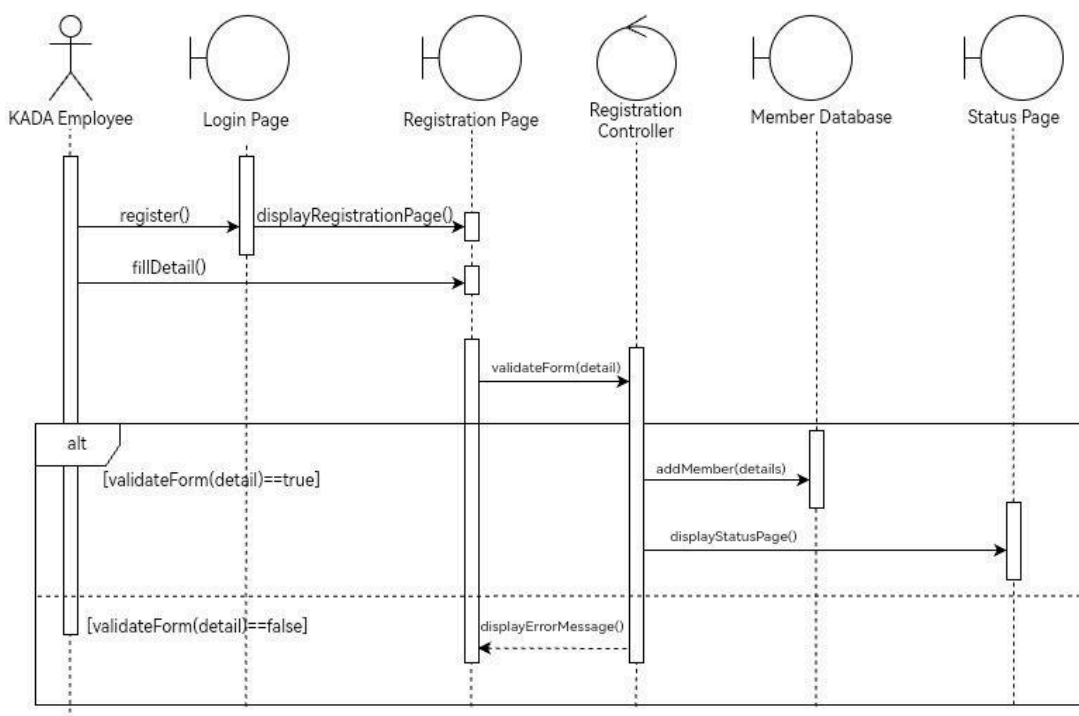


Figure 1.5: Sequence Diagram for Apply for Membership

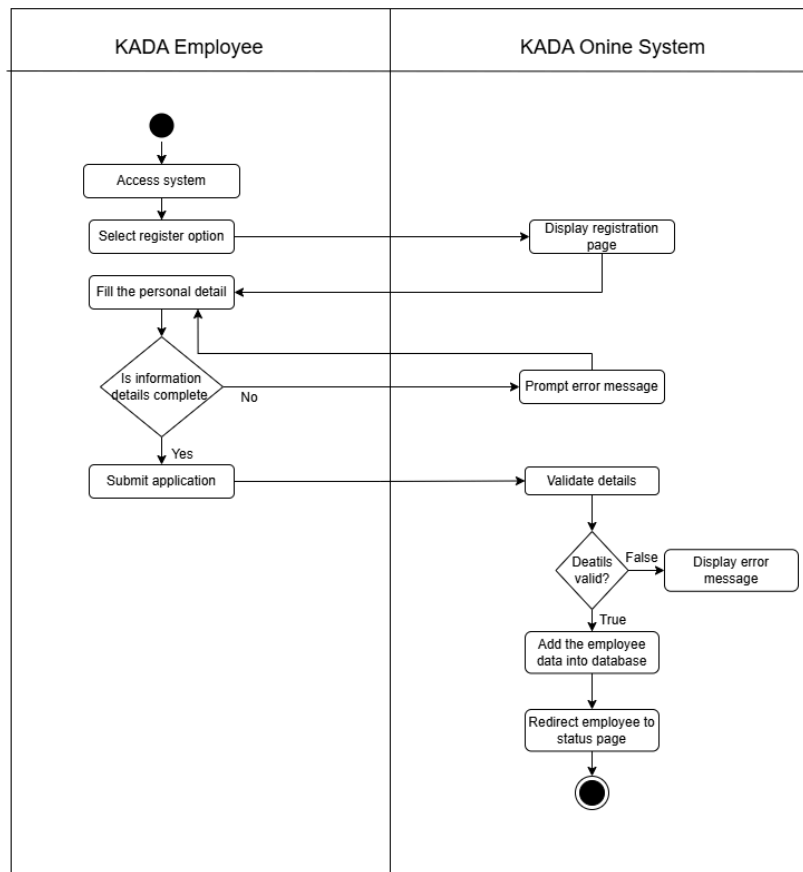


Figure1.6: Activity Diagram for Apply for Membership

2.4.2 US002 User Story Apply for Loan

Table 2.2: User Story Description for Apply for Loan

User story: Apply for a Loan
ID: US002
User Story Description: As a cooperative member I want to apply for a loan from the cooperative So that I can get the
Flow of events: 1. The member logs in to the system by using the registered email. 2. The member chooses the “Permohonan Pembiayaan Anggota ” on the navigation bar. 3. The member fills in all the personal details in the form. 4. The member needs to upload the required documents such as "Butir-Butir Penjamin" and "Borang Pengesahan Majikan". 5. The member presses the submit button to submit the application form.

6. The system will prompt a message to confirm the submission.
Alternative flow: 1. If the personal information is incomplete, the system will prompt a message to remind the employee to fill in the incomplete place. 2. If the application is rejected, the employee must reapply and cannot log in to the system.
Acceptance Criteria Precondition: 1. The member logs in to the system. Postcondition: 1. If the application is approved, the member will receive a notification and the employee can log into the system to enjoy the benefits provided by the cooperative. 2. If the application is rejected, the employee will receive a notification. The employee must reapply and cannot log in to the system. Other conditions: 1. A rejected application must clearly notify the reason for the rejection so that the employee can easily reapply.
Exception flow: 1. If the system is down, the system will prompt a message to advise the employee to try later. 2. If the employee has invalid details, the system will prompt a message to remind the employee to re-enter

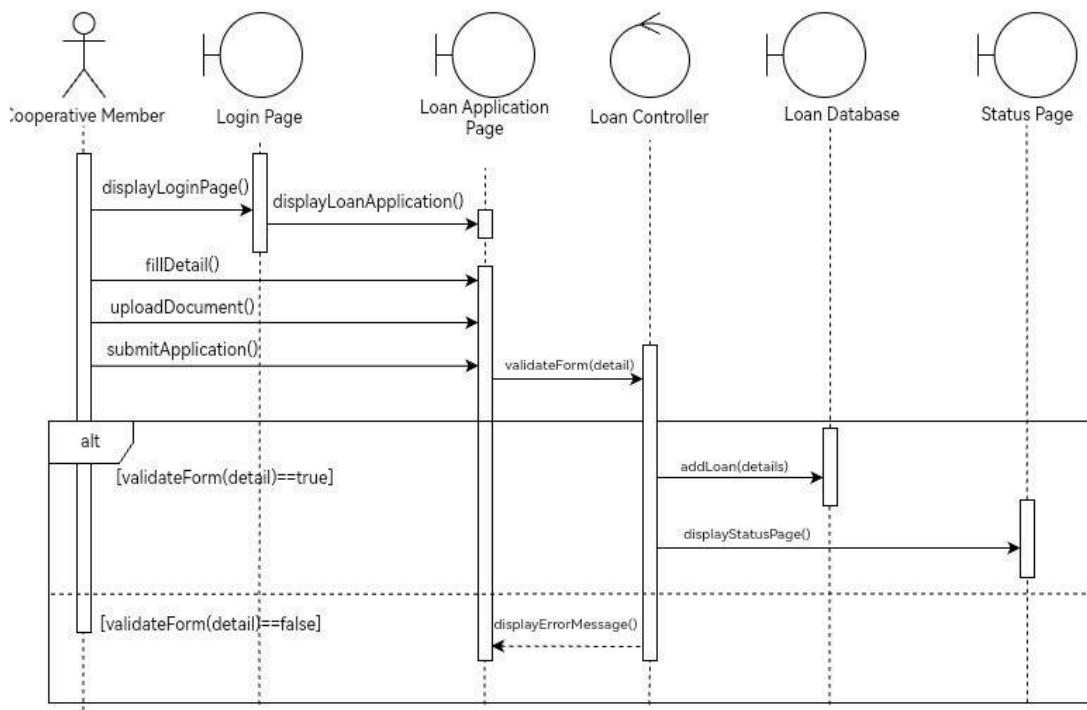


Figure 1.7: Sequence Diagram for Apply for Loan

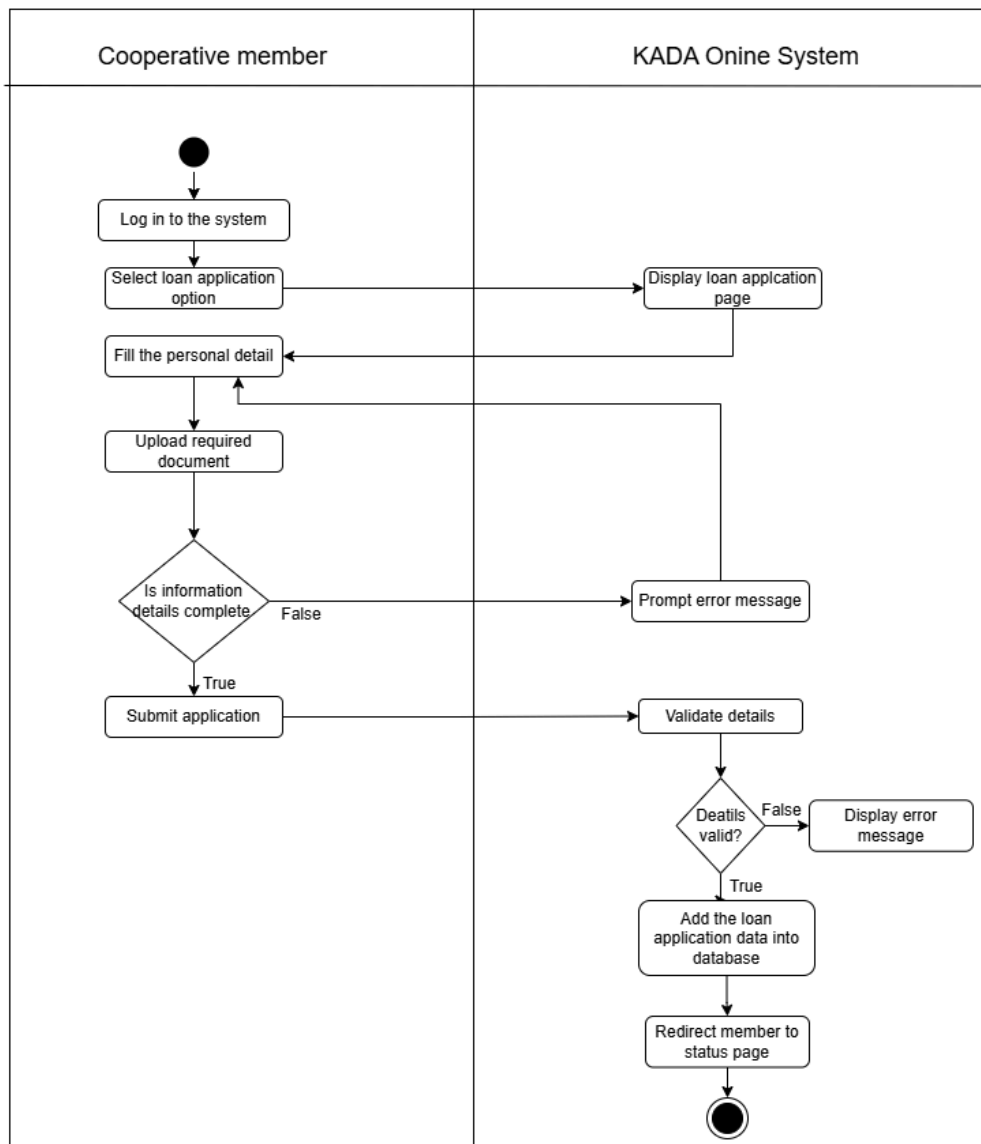


Figure 1.8: Activity Diagram for Apply for Loan

2.4.3 US003: User Story View Financial Status

Table 2.2: User Story Description for View Financial Status

User story: View Financial Status
ID: US003
User Story Description: As a cooperative member I want to view my financial status So that I can get to know my current financial status

Flow of events: 1. The member logs in to the system by using the registered email. 2. The member navigates to the “Profile” section. 3. The member reviews the financial status.
Alternative flow: -
Acceptance Criteria Precondition: 1. The member logs in to the system. Postcondition: 1. The member successfully views the financial information. Other conditions: -
Exception flow: 1. If the system is down, the system will prompt a message to advise the employee to try later.

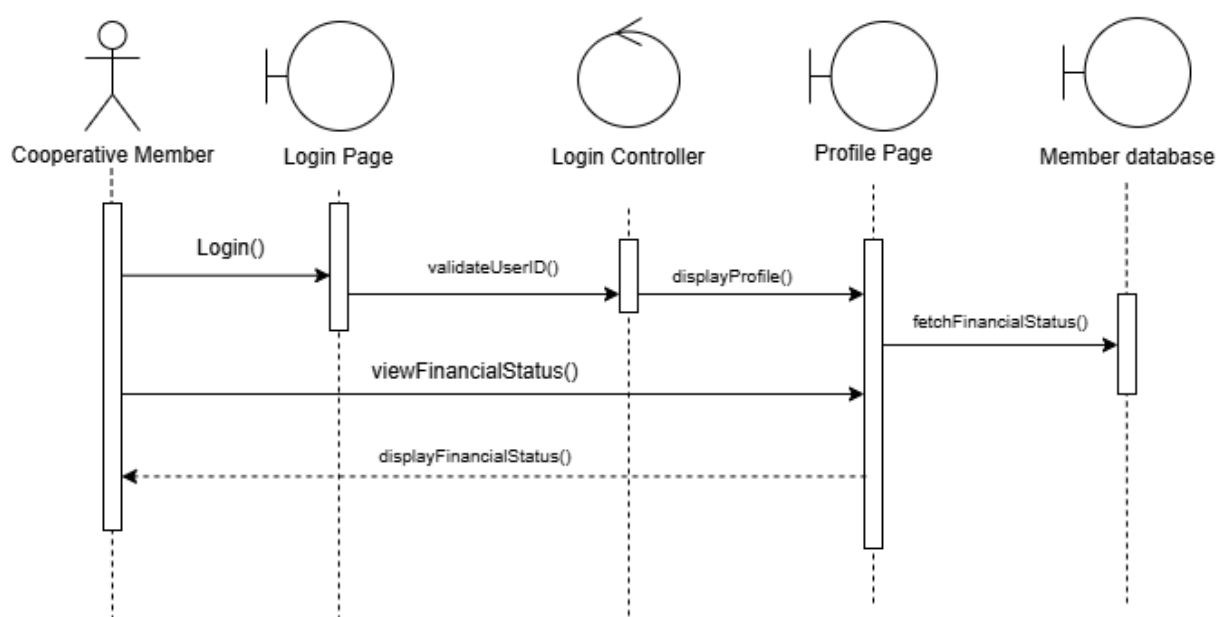


Figure 1.9: Sequence Diagram for View Financial Status

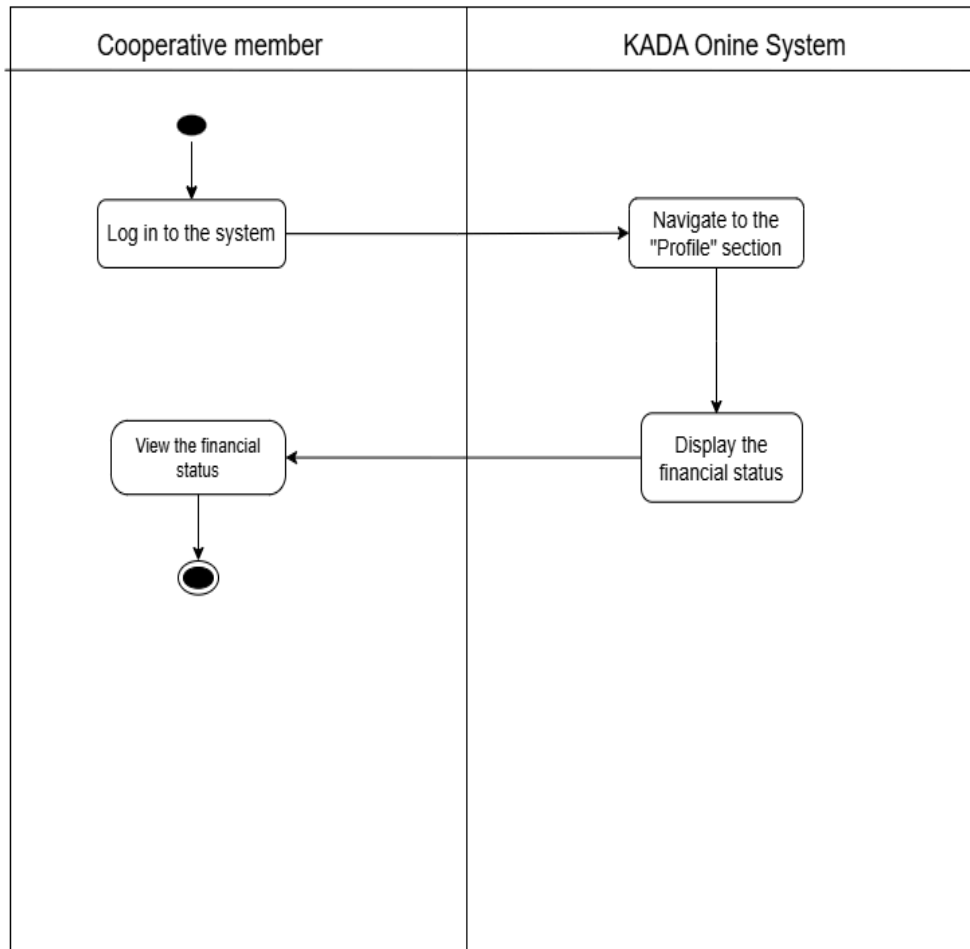


Figure2.0: Activity Diagram for View Financial Status

2.4.4 US004: User Story <User Story 4>

Table 2.4: User Story Description for <Validate membership registration and loan application>

User story: Process membership registration and loan application
ID: US04
User Story Description As a admin or as a board director, I want to process the membership registration and loan application So that new members can join and members can access loans.
Flow of events: - Admin logs into the system. - Admin selects the "Membership Applications" or "Loan Applications" option. - The system retrieves and displays pending applications. - Admin reviews the details of an application. - Decision Point: If Membership Application: - Admin approves or rejects the membership request. - The system updates the database and sends a notification to the applicant (via email or dashboard). If Loan Application: - Admin reviews loan eligibility and supporting documents. - Admin approves or rejects the loan application. - The system updates the database and sends a notification to the applicant.
Alternative flow <i>n</i>: If the application details are incomplete, the system prompts the admin to request missing information from the applicant.
Acceptance Criteria Precondition: - Admin logged in to the system. Postcondition: - The status of membership or loan application is updated in the system. - Notifications are sent to applicants within 2 minutes of status change.
Exception flow: -

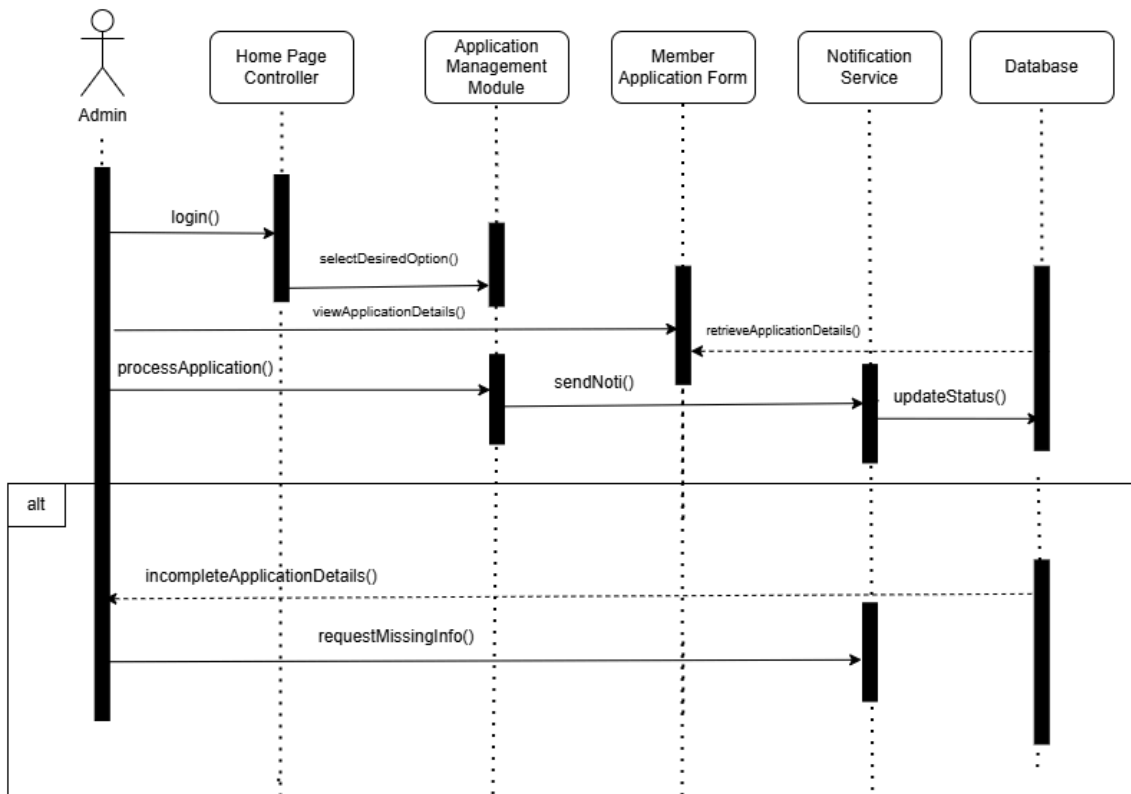


Figure 2.1 : Sequence Diagram for Process Membership Registration and Loan Application

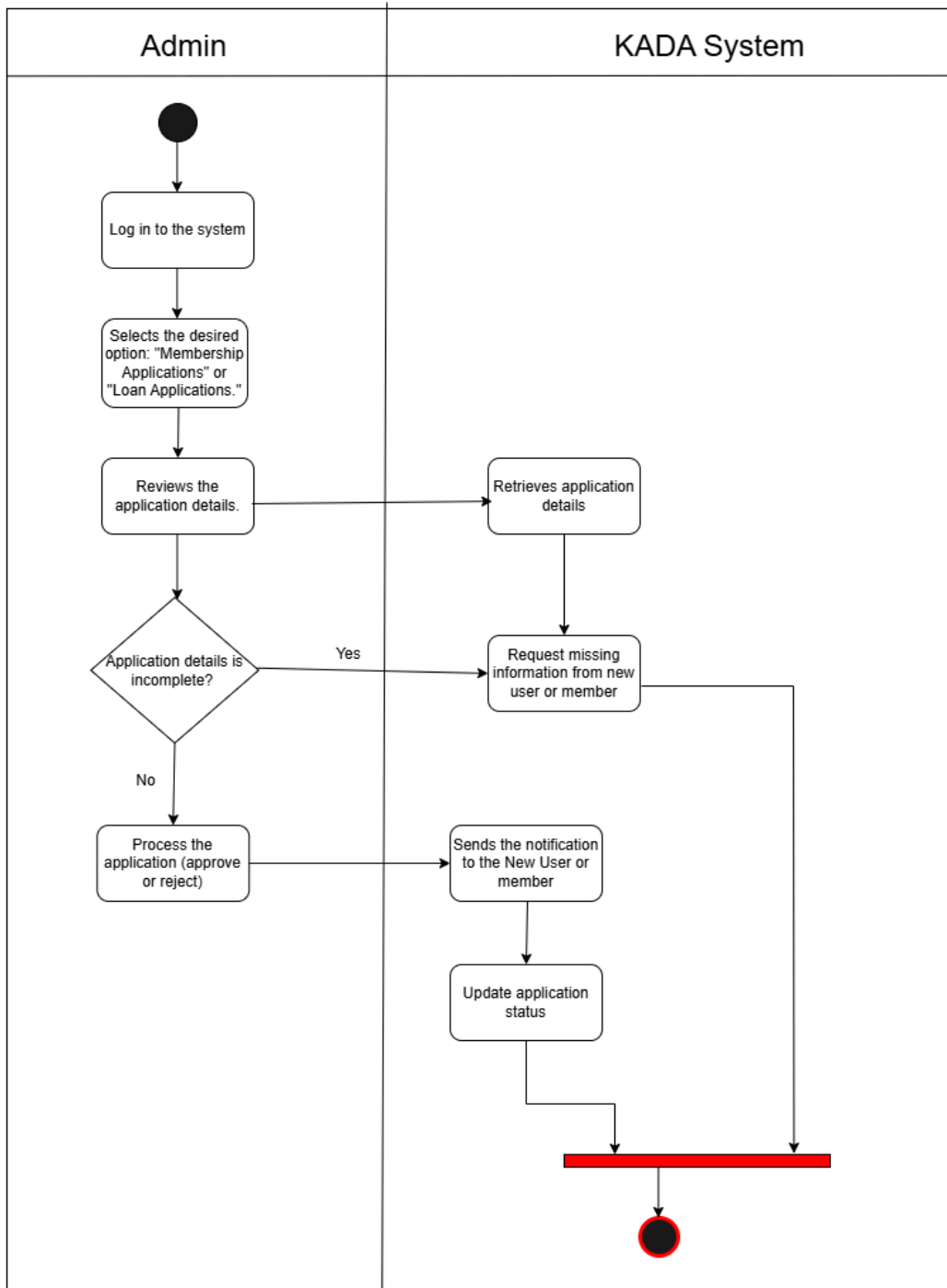


Figure 2.2: Activity Diagram for Validate Membership Registration and Loan Application

2.4.5 US005: User Story <User Story 5>

Table 2.5: User Story Description for <Process Reports>

User story: Process Reports
ID: US05
User Story Description As an admin , I want to process reports So that I can monitor the cooperative's performance.
Flow of events: 1. The Admin logs into the system. 2. The Admin selects the report type, either monthly or annually. 3. The system retrieves relevant data from the database, including New member applications, Loan applications, Fees claimed, and Contributions claimed. 4. The system generates a financial summary and detailed breakdown of each category for the specified period. 5. The system displays the report, providing options to: view the report in detail, export the report (PDF, Excel) or print the report.
Alternative flow <i>n</i>: 1. If an invalid format is selected, the system prompts the user to select the correct format.
Acceptance Criteria Precondition: 1. The Admin logged in to the system. Postcondition: 1. Accurate and timely reports are generated. 2. Visual charts and summaries are easy to understand. 3. Exportable to formats like PDF or Excel.
Exception flow: -

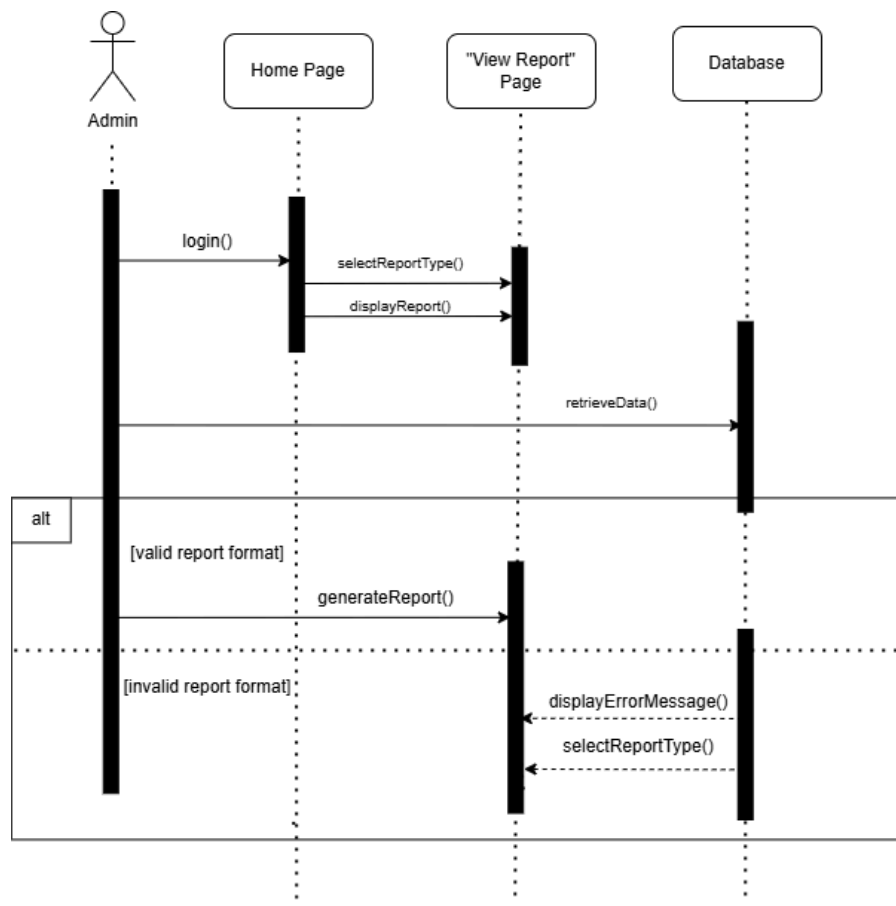


Figure 2.3: Sequence Diagram for Process Reports

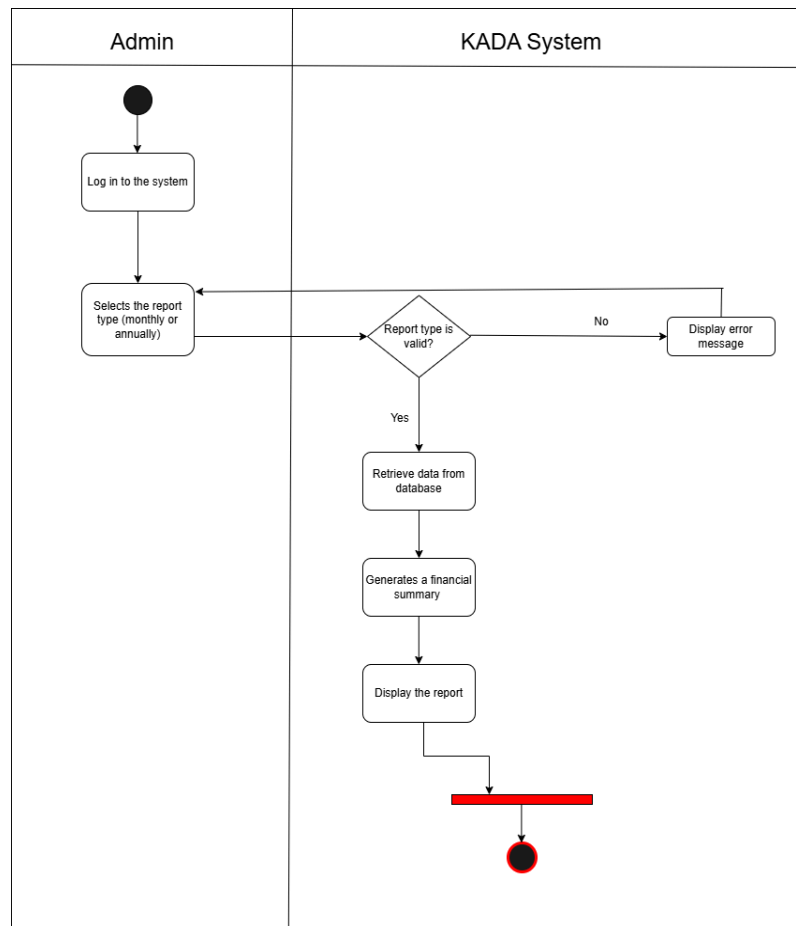


Figure 2.4: Activity Diagram for Process Reports

2.5 Performance and Other Requirements

Software System Attributes:

- Usability: The user-friendly system can ease the user to interact with the system which provides clear instructions and navigation.
- Reliability: The system will be able to recover from server downtime within 30 minutes to minimize disruption.
- Security: The system will implement two-factor authentication (2FA) for login and further actions such as loan approvals.
- Maintainability: The system is easy to update or modify to individual components without affecting the entire system.
- Compatibility: The system is compatible with major web browsers such as Chrome, Safari, Firefox, iOS and Android.
- Portability: The system can run on all desktop platforms and mobile devices and also deployable on different cloud platforms.
- Safety: The system is able to ensure that data cannot be accidentally overwritten, deleted or corrupted during user transactions or system operations.
- Legal and Regulatory: The system meets relevant certifications for data handling such as ISO 27001 and any third-party software used adheres to licensing agreements.
- Environmental: The system shall help KADA significantly to reduce its reliance on paper-based documentation by digitizing registration and loan application processes.

Performance:

- Response Time: The system shall process member registration and loan applications within 2 seconds under normal load conditions.
- Availability: The system shall maintain 99.5% uptime during business hours and 99% uptime overall including scheduled maintenance periods.
- Capacity: The system shall support up to 10,000 registered members' data at launch and scale to accommodate 50,000 members over the next 5 years.
- Throughput: The system shall process at least 50 registration requests per minute and 30 loan applications per minute both under normal load conditions.

2.6 Design Constraints

Environmental constraints: The system must function in an environment with variable internet speeds to ensure usability in both urban and rural areas with limited network infrastructure.

Hardware constraints: The system must be designed to operate on devices with basic hardware capabilities such as mobile devices with at least 2GB of RAM and must be deployed on a server with a minimum configuration of 16GB RAM, 8-core CPU and 1TB SSD storage.

Security constraints: The system must comply with Personal Data Protection Act (PDPA) requirements and encrypt all data transmission between client and server using SSL/TLS protocols.

Compatibility constraints: The system must be compatible with modern web browsers such as Chrome, Firefox and Safari and both Android and iOS devices for mobile apps.

Performance constraints: The system must handle up to 1,000 concurrent users accessing it during peak periods within 2 seconds to perform actions such as member registration and loan application.